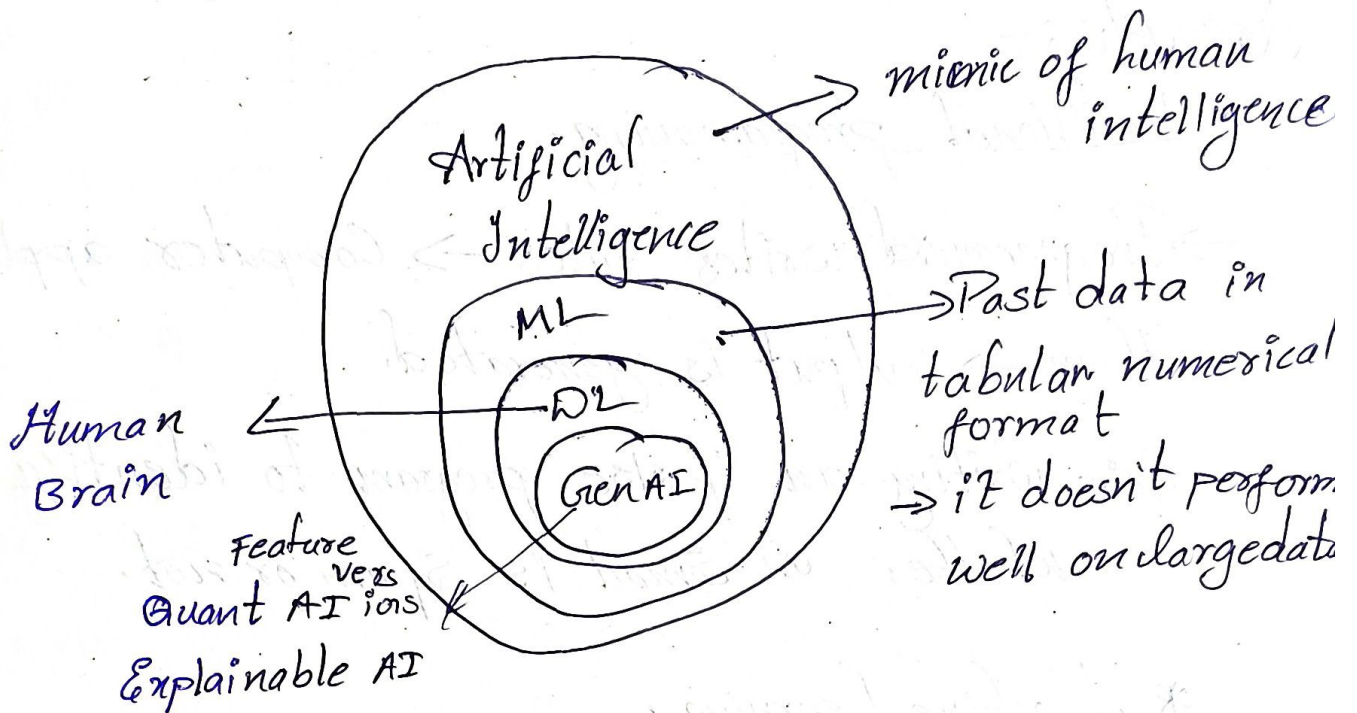


12/11/25

Day-1

Machine Learning:-



For an IQ major things

- ⇒ Observation
- ⇒ Analyzing
- ⇒ Thinking
- ⇒ Prediction

What is Machine Learning:-

ML is a branch or subset of A.I. that enables ^{the} Computer to learn from data and that make predictions (or) decisions without being explicitly programmed.

In simple words:-

Instead of writing a code to solve a specific problem, we feed the machine with

data and examples and it learns patterns from them.

Example:-

* Traditional programming:-

→ Programmer writes rules → Computer applies them → output is generated.

Ex:- writing an if-else program to identify whether an email is spam or not.

* Machine Learning:-

→ We give the computer thousands of emails labeled "spam" or "not spam" →

the computer learns the rules automatically

→ It predicts if a new email is a spam.

Key Components of Machine Learning:-

→ Data:- The information the machine learns from

→ Features:- Input variables used to make predictions

→ Model :- The algorithm that learns from data.

→ Training :- The process of teaching the ML model using data.

→ Prediction :- using the trained model on new unseen data.

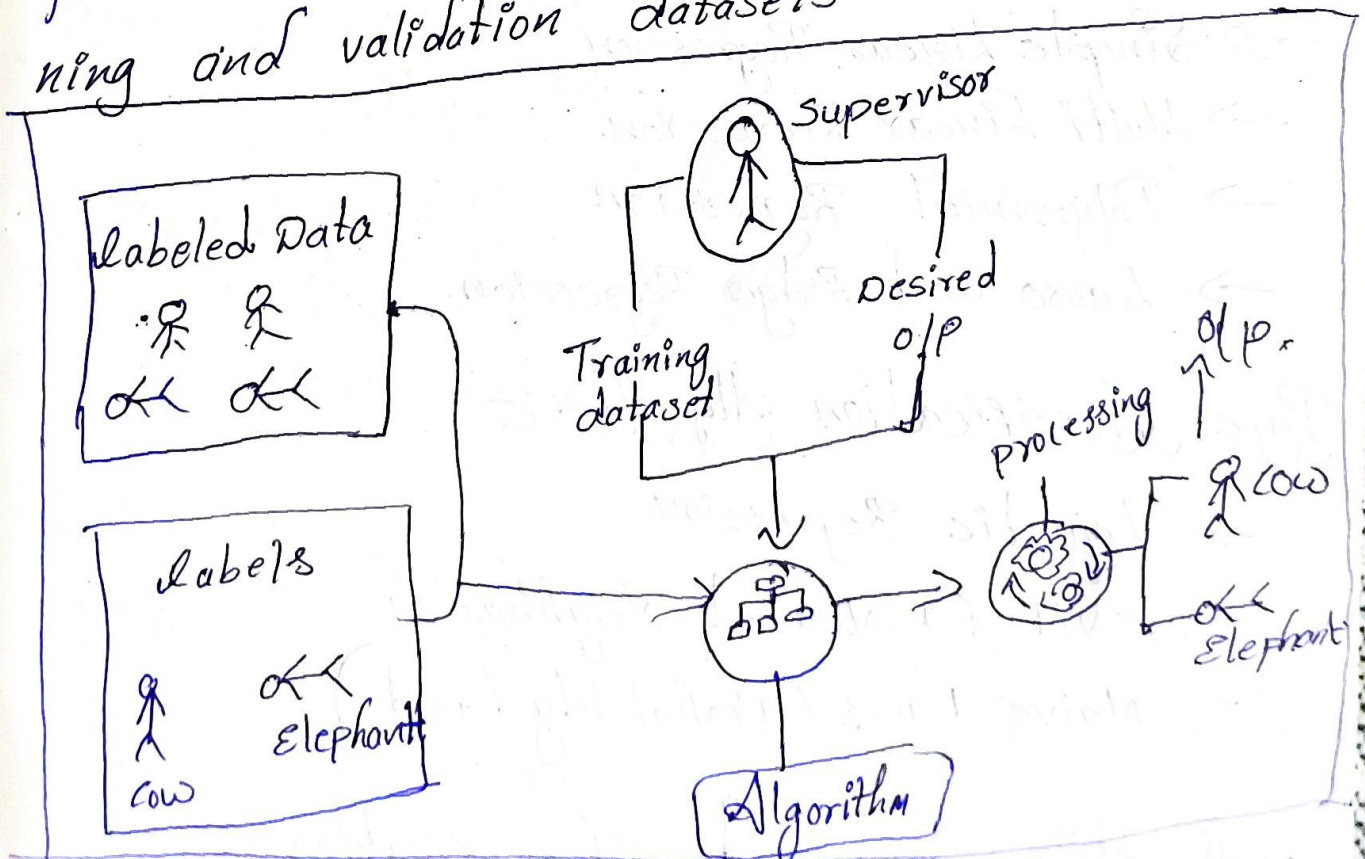
Types of Machine Learning:-

- Supervised Learning - Labeled data
- Unsupervised Learning - unlabeled data
- Reinforcement Learning -

Supervised Learning:- It is defined as when a model gets trained on a "Labeled Dataset".

Labeled datasets have both input & o/p parameters.

In supervised Learning Algorithms learns to map points b/w i/p & current o/p. It has both training and validation datasets labeled.

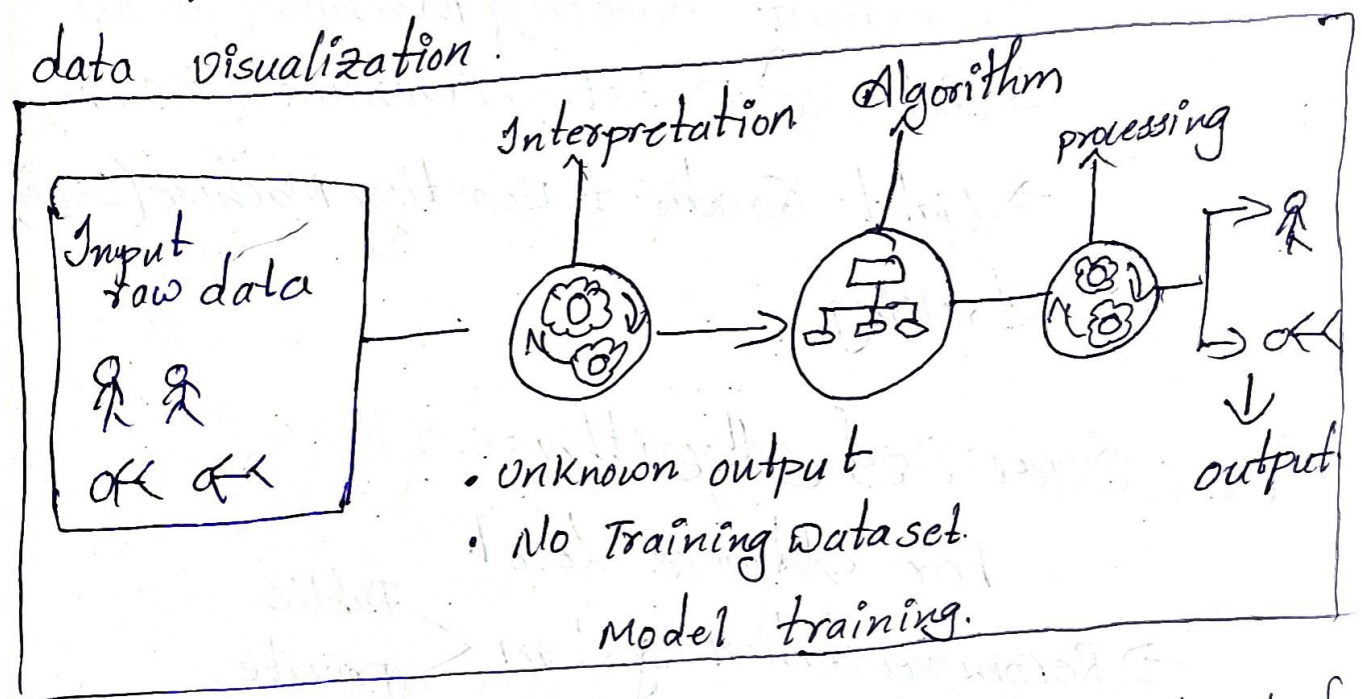


Ex:- Image, speech & text, Healthcare & finance

Day-02

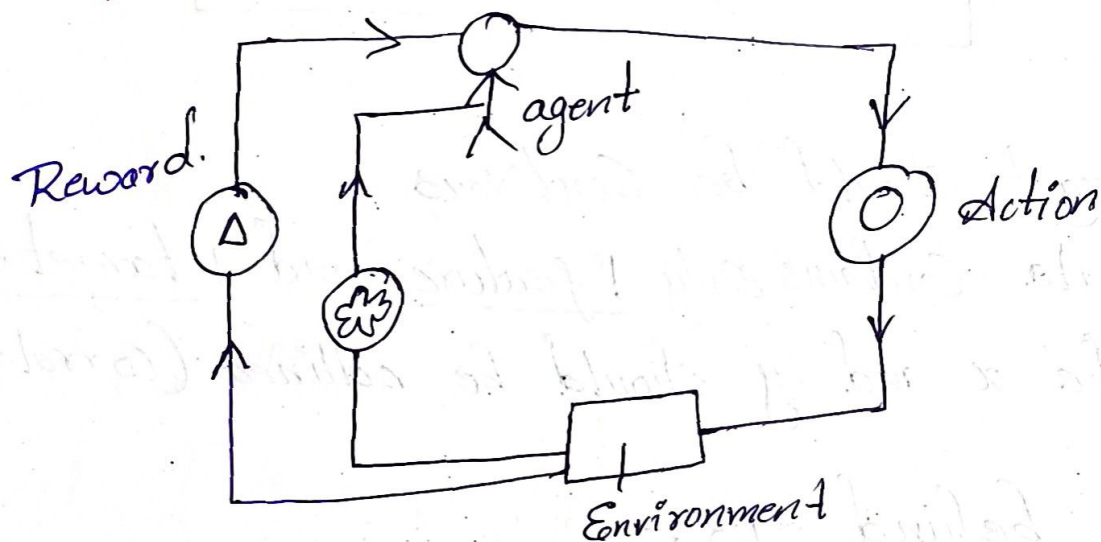
Simple Linear Regression:-

Unsupervised Learning: It works with unlabeled data, meaning there are no predefined outputs. The algorithm finds hidden patterns, groups or relationships within the data on its own. It's mainly used for clustering, dimensionality reduction and data visualization.

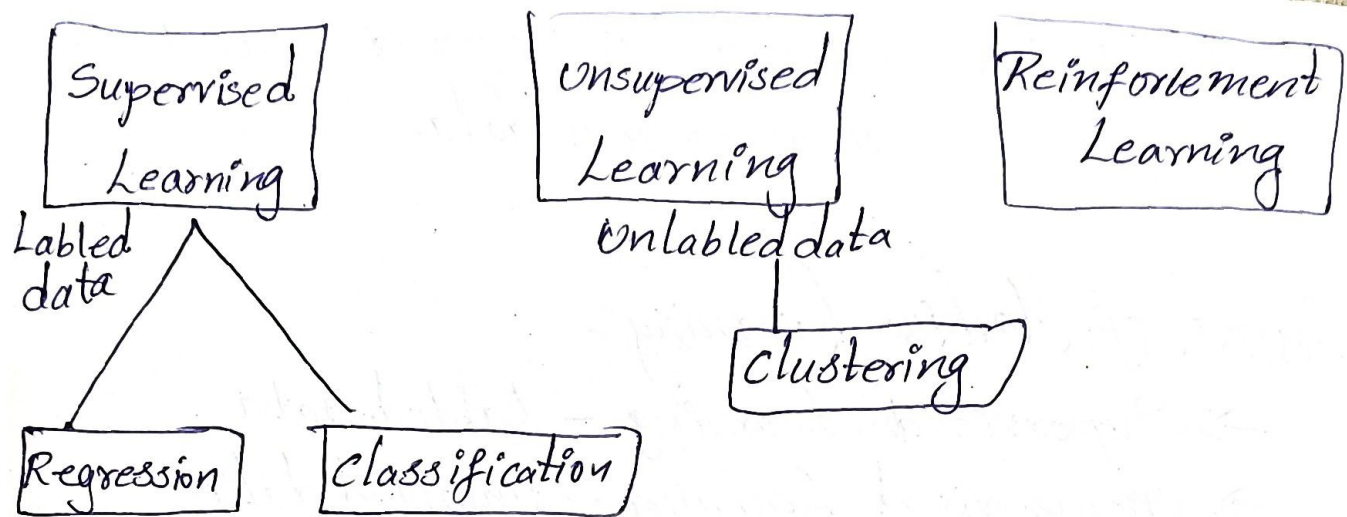


- It uses when data is unlabeled or unstructured.
- useful for Exploratory analysis, clustering or feature Extension.
- Common in marketing, recommendation systems and fraud detection where patterns matter more than labels.

Reinforcement Learning:- It trains an agent to make a sequence of decisions through trial and error. The agent interacts with the Environment, receives feedback in the form of rewards or penalties and learns optimal actions over time.



- When you need an agent to learn by interacting with an Environment.
- Best for decision-making or optimization tasks involving trial and feedback loops.
- Used when long-term performance or adaptive behaviour is more important than immediate accuracy.



if prediction point is continuous we use Regression.

Ex:- MPG, BMI

if prediction point is categorical / classes we use

classification

Ex:- Y/N, T/F, S/R/W

Pure Regression Algorithms:-

- Simple Linear Regression
- Multi Linear Regression
- Polynomial Regression
- Lasso and Ridge Regression

Pure Classification Algorithm:-

- Logistic Regression.
- KNN (K-Nearest Neighbours)
- Naive Bayes (Probability based)

Both Regression and Classification:-

- Decision Tree Algorithm. $\leftarrow \begin{matrix} R \\ C \end{matrix}$
- SVM (Support Vector Machine) Algorithm. $\leftarrow \begin{matrix} R \\ C \end{matrix}$

Ensemble Techniques:-

→ Bagging.

Random Forest $\leftarrow \begin{matrix} R \\ C \end{matrix}$

→ Boosting

→ Adaptive Boosting Machine (AdaBoost)

→ Gradient Boosting Machine (GBM)

→ Extreme $\uparrow$ Boosting Machine (XGBM)

→ Light Gradient Boosting Machine (LGBM)

→ Stacking.

Semi-Supervised Algorithms:-

For Advance Level.

→ Recommendation system $\leftarrow \begin{matrix} \text{Public} \\ \text{Private} \end{matrix}$

→ Time series Analysis

→ Anomaly.

→ SL:-

• When you labeled data & want to predict outcomes

• Ideal for classification (like spam detection) or regression tasks (like price forecasting).

• Best used in domains where historical data with outcomes is already available.